Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building **AI job orchestration**, LLM inference routing, **data pipelines**, and Python backend services in production, owning features from requirements through release, failure recovery, and debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built fault-tolerant **AI job orchestration** with FastAPI workers, explicit job-state workflows, and failure isolation, processing 1K+ LLM jobs at 3x throughput without cascading failures under concurrent load.
- Designed multi-tenant storage with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms, and **monitored** integration jobs to sustain 99%+ reliability through CI/CD releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **model routing** layer across OpenAI, Anthropic, and Qwen inference APIs with dynamic switching, timeouts, and fallbacks, serving 10K+ daily requests for 150+ instructors on AWS.
- Designed a **multithreaded batch pipeline** with secure file ingestion for PDF, DOCX, and ZIP uploads, processing 1K+ files per batch with fault-tolerant execution and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed **REST API** endpoints with pagination, field filtering, and caching for clients on low-bandwidth networks, improving response times 30% and cutting payload sizes across large-scale integrations used by distributed clients.
- Wrote 20+ automated unit and integration test suites with PyTest and Mockito, run nightly through Jenkins and GitHub-triggered branches across Windows, Linux, and macOS in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **distributed** Python FastAPI microservices for authentication with role-based access, billing, and customer records, syncing 2M+ records across locations through real-time WebSocket updates for customer and lead workflows.
- Shipped through Jenkins CI/CD with Docker and staged releases, and debugged production issues with Sentry, Azure Monitor, and Kibana, cutting deployment time 50% and reaching 95%+ successful build validation.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built customer-facing Flask APIs for bookings, pricing, and coupons with configurable eligibility rules, keeping responses under 200ms across 50K+ records with Firebase real-time updates on Android, iOS, and web.
- Integrated multi-currency payment providers through idempotent webhooks for refunds and disputes, cutting manual handling about 70%, and implemented wallet-signature login that removed password storage for wallet users.

Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, Bash
Backend: FastAPI, Flask, Node.js, REST APIs, Microservices, Distributed Systems, Background Jobs, Webhooks
AI & Data: LLM APIs, AI Agents, Model Routing, LLM Evaluation, Embeddings, FAISS, PyTorch, ETL Pipelines
Infra: AWS, GCP, Docker, Kubernetes, PostgreSQL, Redis, CI/CD, GitHub Actions, Sentry, Linux
Practices: Git, Debugging, Code Reviews, PyTest, Performance Tuning, AI Coding Tools, Documentation, Agile

Projects

Distributed LLM Evaluation Harness | *Python, Celery, Redis, PostgreSQL, Docker*

- Designed a harness that shards **evaluation** tasks across worker queues, runs model and agent outputs against graders, retries failed jobs, and stores scores and traces for **dataset** review.

Durable AI Agent Workflow Engine | *Python, FastAPI, PostgreSQL, Kubernetes*

- Designed an engine for **long-running agent** workflows that checkpoints each step to Postgres, resumes after crashes, limits concurrent tool calls, and lets domain experts review and approve generated outputs.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built an **AI application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy in production.